

Date _____

Question 06:

Pseudocode:

START

DECLARE current SOC, Req SOC, duration, Time, Req-Charge, Charge-Cost, Park-Cost,
Total-Discout, Total Payable, Base-Rate, Base-Park, Discount-P, Rate-Charge-Cost,
Charge-Discout, Park-Discout: FLOAT DECLARE eligible: BOOLEAN

DECLARE vehicle, member, disabled, available: CHAR

DECLARE Priority, Time-Status, Message: STRING

OUTPUT "Enter vehicle type (E/H): "

INPUT Vtype

OUTPUT "Enter Battery charge level as a percentage: "

INPUT current SOC

OUTPUT "Enter Required charging level as a percentage: "

INPUT Req SOC

OUTPUT "Enter expected parking duration in hours: "

INPUT duration

OUTPUT "Enter Current time in 24-hour format: "

INPUT Time

OUTPUT "Do you have a parking membership (Y/N): "

INPUT member

OUTPUT "Do you have disabled-person priority status? (Y/N): "

INPUT disabled

OUTPUT "Is the charging station currently available? (Y/N): "

INPUT available

chargeC ← 0

ParkC ← 0

Total-discout ← 0

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```
Req_charge ← 0
Priority ← "None"
Time_Status ← "None"
Message ← "", eligible ← TRUE, Req_charge ← Req_SOC - current_SOC
IF available = 'N' THEN
    IF vehicle = 'H' THEN
        OUTPUT "Charging unavailable; Parking only."
    ELSE
        OUTPUT "No charging slots available."
    ENDIF
    eligible ← FALSE
ELSE IF vehicle = 'H' AND current_SOC ≥ 40 THEN
    OUTPUT "Vehicle does not qualify for EV charging."
    eligible ← FALSE
ELSE IF Req_charge ≤ 0 THEN
    OUTPUT "No charging required"
    eligible ← FALSE
ENDIF
IF eligible = TRUE THEN
    IF current_SOC ≤ 15 AND Req_SOC ≥ 80 THEN
        Priority ← "Emergency Charging Priority"
    ELSE IF disabled = 'X' OR (member = 'Y' AND current ≤ 30) THEN
        Priority ← "Priority Charging"
    ELSE
        Priority ← "Normal Charging"
    ENDIF
    Discount_P ← 0.0
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IF Time $\geq$ 17 AND Time $<$ 22 THEN

Time_status $\leftarrow$ "Peak"

Base_Rate $\leftarrow$ 50

IF member = 'Y' AND Priority $\neq$ "Emergency Charging Priority" THEN

Discount_P $\leftarrow$ 0.10

ENDIF

ELSE

Time_status $\leftarrow$ "Off-peak"

Base_Rate $\leftarrow$ 35

IF member = 'Y' AND Priority $\neq$ "Emergency Charging Priority" THEN

Discount_P $\leftarrow$ 0.20

ENDIF

ENDIF

Raw_Charge_Cost $\leftarrow$ Req_charge * Base_Rate

Charge_Discount $\leftarrow$ Raw_Charge_Cost * Discount_P

Charge_Cost $\leftarrow$ Raw_Charge_Cost - Charge_Discount

Total_Discount $\leftarrow$ Total_Discount +

ENDIF

IF duration $\leq$ 2 THEN

Base_Park = 200

ELSEIF duration $\leq$ 5 THEN

Base_Park = 400

ELSE

Base_Park = 700

ENDIF

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```
IF disabled = 'Y' THEN
    Park_Cost = 0
    Total_Discount ← Total_Discount + Base_Park
ELSE IF member = 'Y' THEN
    Park_Discount ← Base_Park * 0.20
    Park_Cost ← Base_Park - Park_Discount
    Total_Discount ← Total_Discount + Park_Discount
ELSE
    Park_Cost ← Base_Park
ENDIF
IF duration > 8 THEN
    Message ← "Long stay warning... Please relocate your vehicle after charging."
ELSE
    Message ← "Standard parking duration"
ENDIF
Total_Payable ← Charge_Cost + Park_Cost

OUTPUT "Vehicle type:" vehicle
OUTPUT "Current Battery:" current_SoR | Required Battery:" Req_SoR
OUTPUT "Charging priority:" Priority
OUTPUT "Time status:" Time_Status
OUTPUT "Charging Cost:" Charge_Cost
OUTPUT "Parking Cost:" Park_Cost
OUTPUT "Total Discounts applied:" Total_Discount
OUTPUT "Final amount:" Total_Payable
OUTPUT Message
END
```

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IPD

INPUT	Process	OUTPUT
- v type (Vehicle type: (E/H)) - current SOC (current battery level %) - Req. SOC (Required battery level %) - duration (Expected parking duration in hours) - Time (current time in 24-hour format) - member (Parking membership: Y/N) - disabled (Disabled-person priority: Y/N) - available (charging station availability: Y/N)	(1) Eligibility & Setup - calculate Req. Charge ($\text{Req. SOC} - \text{current SOC}$) - Check if charging is available, if vehicle qualifies (Hybrid with $\geq 40\%$ SOC don't) and if charge is needed ($\text{Req. Charge} > 0$). Set eligible flag (2) Determine Priority (Emergency, Priority, or Normal) based on, current SOC, Req. SOC, disabled, and member status. (3) Determine Time Status (Peak or Off-Peak) based on Time (Peak is 17 to < 22). Set base Rate (\$5 for peak, \$3 for off-peak). (4) Calculate Raw Charge Cost ($\text{Req. Charge} * \text{Base Rate}$) Apply Discount-P based on member and priority status to calculate Charge - Cost - Determine Base Park fee - Apply parking discounts,	Status Messages - "Charging unavailable" - "No charging slots available" - Vehicle does not qualify for EV charging - No charging required Final Receipt Outputs - Vehicle type - Current Battery & Required Battery - Time status - Charging Cost - Parking cost - Total discounts applied - Final amount = - Total Payable - Message (Long stay warning or direct)

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PAC

Data Given	Processing Required	Required Results.
Vehicle type (vtype)	(1) Calculate required charge amount	- early exit messages for ineligibility.
- Current SOC (current_SOC)	(2) Validate charging eligibility	-> Req. charge
- Required SOC (Req_SOC)	(3) Assign charging priority	-> Priority
- Parking duration (duration)	(4) Determine base charging rate based on time of day	- Time_Status string
- Current time (Time)	(5) Calculate charging and net charging discount	- net Charge_Lost
- Membership status (member)	(6) Determine base parking rate based on duration.	- net Park_Lost
- Disabled status (disabled)	(7) Accumulate total discount	- Total_Discount
- Station availability (available)	(8) Calculate final payable	- Total_Payable
	(9) Generating parking duration message	- Duration Message

Question 06

START



OUTPUT "Enter vehicle type (E/H):"

INPUT Vtype

OUTPUT "Enter battery charge level:"

INPUT current SOC

OUTPUT "Enter required charging level:"

INPUT Req-SOC

OUTPUT "Enter expected duration:"

INPUT duration

OUTPUT "Enter current time in 24-hour format:"

INPUT Time

OUTPUT "Parking membership? (Y/N)"

INPUT member

OUTPUT "Do you have disabled person priority?"

INPUT disabled

OUTPUT "Is the charging station available?"

INPUT available



Charge_Cost ← 0

Park_Cost ← 0

Total_Discount ← 0

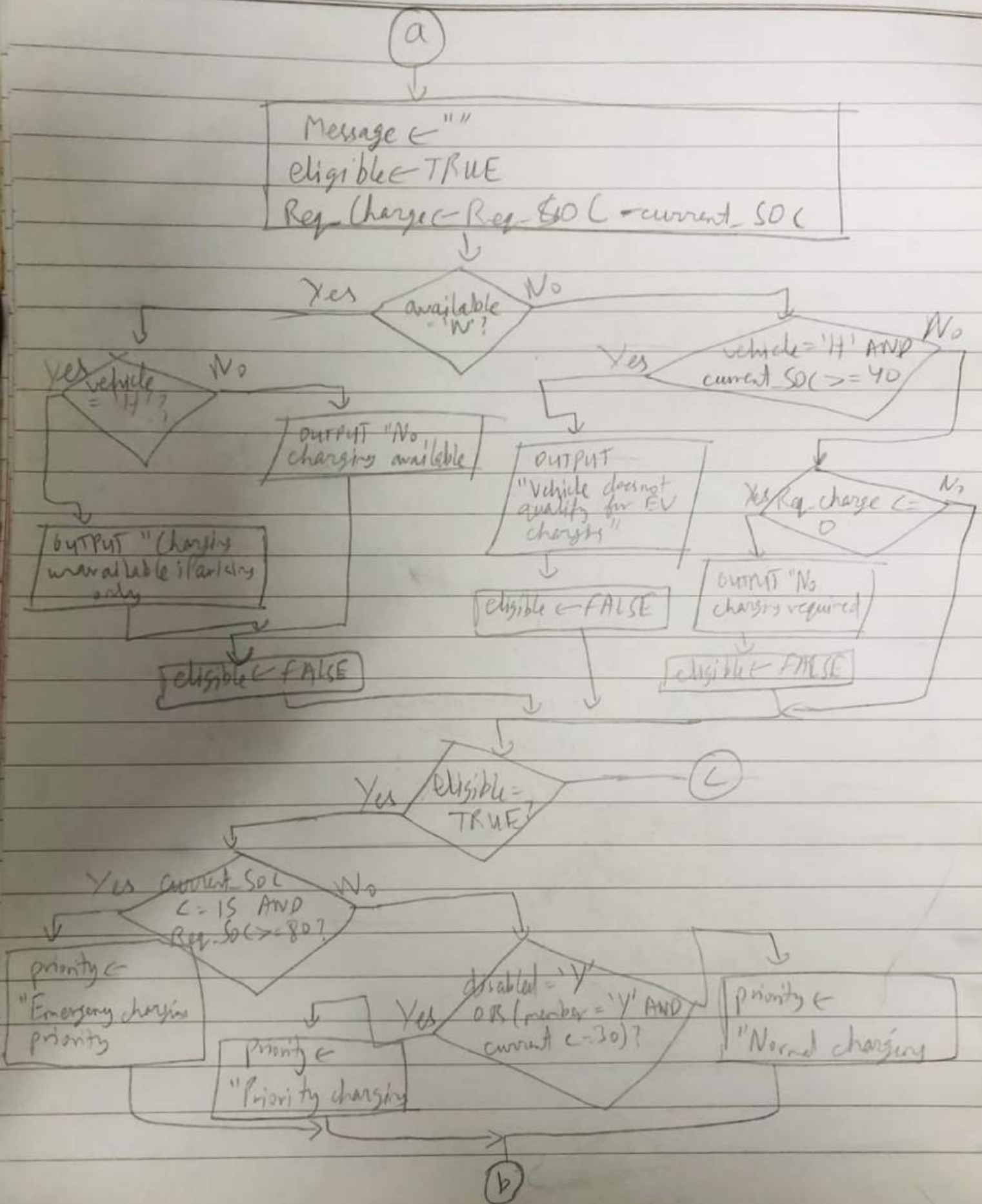
Req_charge ← 0

Priority ← "None"

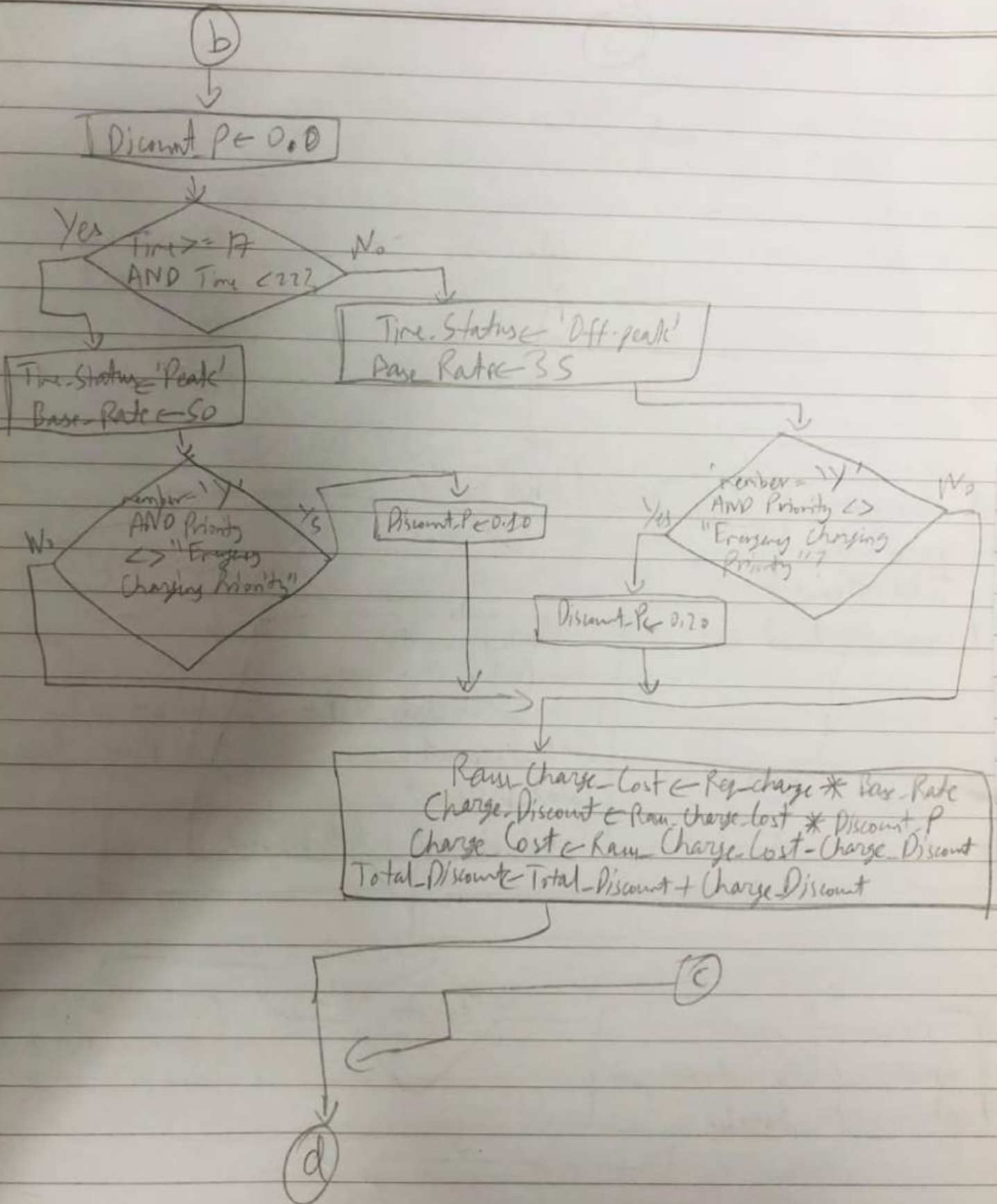
Time_Status ← "None"



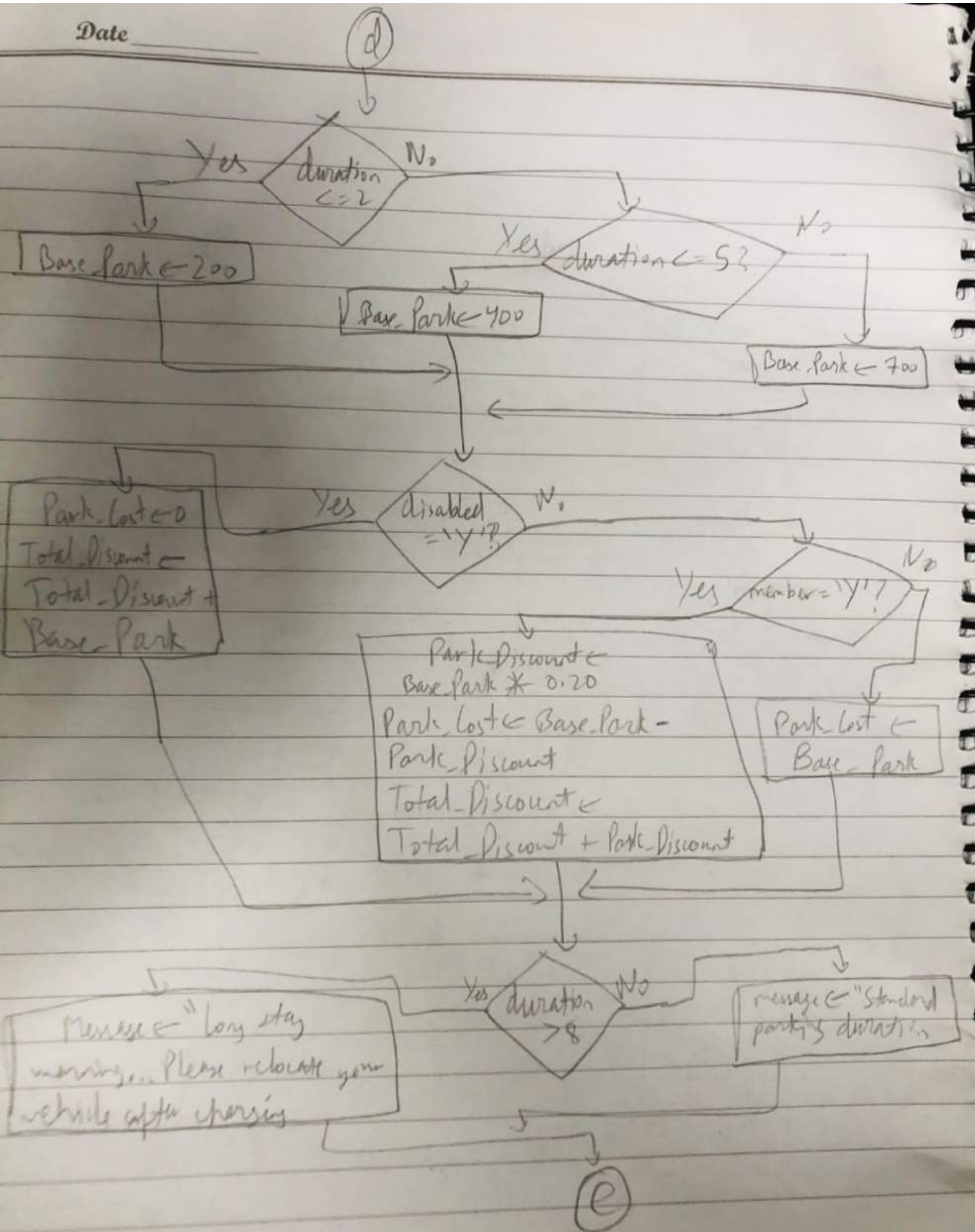
a



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(e)



Total Payable = Charge Cost + Park Cost



OUTPUT "Vehicle type:" vehicle
OUTPUT "Current Battery:" current SOC | Required
Battery:" Req SOC
OUTPUT "Charging priority:" priority
OUTPUT "Time status:" Time Status
OUTPUT "Charging Cost:" Charge Cost
OUTPUT "Parking Cost:" Park Cost
OUTPUT "Total Discounts applied:" Total Discount
OUTPUT "Final amount:" Total Payable
OUTPUT Message



END